

M2 · Random Forest

Model 3 of the ladder · does non-linearity matter? · AI Credit Risk / Loan Approval Engine

In one sentence: a Random Forest grows hundreds of decision trees, each on a random sample of loans and looking at a random subset of features at every split. A leaf's PD is the default rate of the training loans that landed in it, and the forest's PD is the **average over all trees**. Averaging many different, noisy trees cancels their individual mistakes, and the trees find interactions like "high DTI only matters when income is low" on their own.

How it was set up

- Same feature sets as M1 for a fair fight: **A** 189 features, **B** 147 (no LendingClub grades; deployable). Trees need **no scaling or log transforms**.
- **Run on Kaggle** (private CPU notebook, 4 cores, about 30 GB RAM, scikit-learn 1.6.1), driven from the Kaggle CLI, after the laptop run hit its memory limit. Only train and validation data were uploaded, so the test set stayed sealed. Each tree trains on a random half of the loans (*max_samples=0.5*).
- Tuning: 7 settings per version at 200 trees (29 minutes in total), chosen on validation ROC-AUC; then the winner was refit with 500 trees.

Parameters

Parameter	Version A	Version B	What it means
n_estimators	500	500	Number of trees (200 during tuning). More trees = more stable average, never more overfitting.
min_samples_leaf	50	20	Minimum loans per leaf. Larger = smoother, more reliable leaf default rates. Tried 20 / 50 / 100.
max_features	0.3	sqrt	Features considered at each split (sqrt $\approx$ 12-14). Fewer = more diverse trees. Tried sqrt and 0.3.
class_weight	None	None	Up-weighting defaults was checked and not kept.
max_samples, bootstrap, criterion, random_state	0.5, True, gini, 42	0.5, True, gini, 42	Each tree sees a random half of the loans; gini split quality; fixed seed for reproducibility.
Tree size (average)	4,445 nodes, depth 28	10,592 nodes, depth 31	Final fit time: 8.4 min (A), 2.5 min (B).

What the tuning showed (validation ROC-AUC, 200 trees)

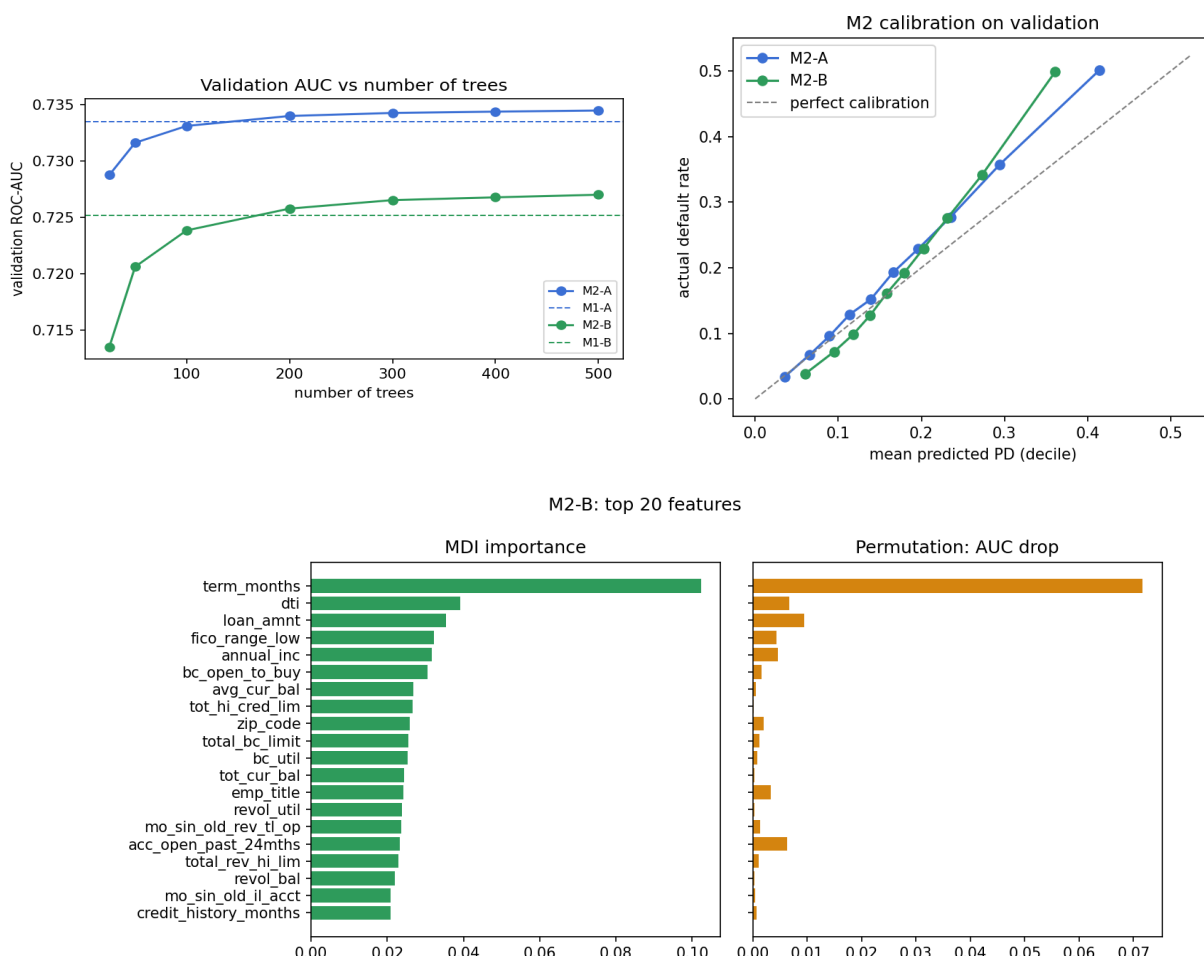
Experiment	Version A	Version B
min_samples_leaf 20 / 50 / 100 (best of max_features)	0.7335 / 0.7340 / 0.7332	0.7258 / 0.7251 / 0.7237
max_features sqrt / 0.3 (best of leaf sizes)	0.7335 / 0.7340	0.7258 / 0.7256
class_weight None vs balanced_subsample: AUC	0.7340 vs 0.7337	0.7258 vs 0.7242
... and Brier (lower is better)	0.1444 vs 0.1869	0.1468 vs 0.1842
Trees 25 $\rightarrow$ 100 $\rightarrow$ 500 (final model)	0.7288 $\rightarrow$ 0.7331 $\rightarrow$ 0.7345	0.7135 $\rightarrow$ 0.7239 $\rightarrow$ 0.7270

Results (validation unless marked)

Model	ROC-AUC	KS	PR-AUC	Brier	Log loss	Mean PD	Actual rate
M0	0.5000	0.0000	0.2035	0.1631	0.5085	17.24%	20.35%
M1-A	0.7335	0.3421	0.4120	0.1445	0.4506	17.73%	20.35%
M2-A	0.7345	0.3410	0.4152	0.1443	0.4503	17.49%	20.35%
M1-B	0.7252	0.3262	0.4045	0.1449	0.4529	18.37%	20.35%
M2-B	0.7270	0.3284	0.4084	0.1467	0.4571	18.17%	20.35%
M2-B (train)	0.8548	0.5517	0.5555	0.1173	0.3767	17.30%	17.24%

The train score is optimistic by design: each training loan was inside the sample of about half the trees. Validation is the honest number.

Verdict: no clear gain. Random Forest changes ROC-AUC by +0.0018 on version B versus Logistic Regression (the bar for a 'clear gain' is +0.005). Averaging independent trees finds little signal that the linear model missed. This does not rule out non-linearity: gradient boosting (M3), where each tree corrects the previous trees' mistakes, is the real test, and M1 remains a strong, explainable candidate.



Key findings

- **M1 vs M2 (validation ROC-AUC):** version A 0.7335 → 0.7345 (+0.0010); version B 0.7252 → 0.7270 (+0.0018). Brier for B: 0.1449 → 0.1467.
- **Approval view (B):** approving the safest 50% gives a bad rate of 9.94% with M2 vs 10.03% with M1 and 20.35% with M0.
- **More trees stop helping early:** B goes from 0.7135 (25 trees) to 0.7239 (100) and 0.7270 (500).
- **What drives the forest (permutation, B):** term_months (+0.0717), loan_amnt (+0.0094), dti (+0.0067), acc_open_past_24mths (+0.0063), annual_inc (+0.0047), fico_range_low (+0.0044). These are the same kinds of drivers M1 found (term, FICO, income, debt), which is a good consistency check.
- **Overfitting shows on train:** M2-B scores ROC-AUC 0.855 on its own training loans but 0.727 on validation (M1-B: 0.711 vs 0.725). Deep trees (average depth 31) memorise training loans; only the validation number counts.
- **Squeezed probabilities:** M2-B over-predicts the safest decile (6.0% predicted vs 3.8% actual) and under-predicts the riskiest (36.1% vs 49.9%). Averaging trees pulls PDs toward the middle, which is why its Brier score is worse than M1's even though its ranking is slightly better.
- **Same ranking, different model:** M1-B and M2-B order applicants almost identically (Spearman 0.937).
- **LendingClub's grades:** A beats B by 0.0075 ROC-AUC, similar to M1.
- **Segments (B):** ROC-AUC 0.689 on 36-month vs 0.672 on 60-month loans; mean PD 18.17% vs actual 20.35%, the same 2015 drift as before.

Importance measures. MDI = how much a feature reduced impurity across all splits (fast but favours continuous features). Permutation = how much validation ROC-AUC drops when that feature is shuffled (20,000-row sample, 3 repeats); this is the more honest measure.